

Implementation

1. Database Setup

Query 1.1: Create and Use Database

```
mysql> CREATE DATABASE StudentDB;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> USE StudentDB;
```

```
Database changed
```

2. Table Creation

Query 2.1: Create Student Table

```
mysql> CREATE TABLE Student (  
->     StudentID INT PRIMARY KEY,  
->     Name      VARCHAR(50),  
->     Email     VARCHAR(50),  
->     Age       INT,  
->     Address   VARCHAR(100)  
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

Query 2.2: Create Instructor Table

```
mysql> CREATE TABLE Instructor (  
->     InstructorID INT PRIMARY KEY,  
->     Name        VARCHAR(50),  
->     Email       VARCHAR(50),  
->     Department  VARCHAR(50)  
-> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

Query 2.3: Create Course Table (with Foreign Key)

```
mysql> CREATE TABLE Course (  
->     CourseID     INT PRIMARY KEY,  
->     CourseName   VARCHAR(50),  
->     Credits      INT,  
->     InstructorID INT,  
->     FOREIGN KEY (InstructorID) REFERENCES Instructor(InstructorID)  
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

Query 2.4: Create Enrollment Table (with Foreign Keys)

```
mysql> CREATE TABLE Enrollment (  
-> EnrollmentID INT PRIMARY KEY,  
-> StudentID INT,  
-> CourseID INT,  
-> EnrollmentDate DATE,  
-> FOREIGN KEY (StudentID) REFERENCES Student(StudentID),  
-> FOREIGN KEY (CourseID) REFERENCES Course(CourseID)  
-> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

3. Inserting Data

Query 3.1: Insert Records into Student Table

```
mysql> INSERT INTO Student VALUES  
-> (1, 'Ramesh', 'ramesh@gmail.com', 20, 'Pune'),  
-> (2, 'Amit', 'amit@gmail.com', 21, 'Mumbai'),  
-> (3, 'Sneha', 'sneha@gmail.com', 19, 'Nashik'),  
-> (4, 'Rahul', 'rahul@gmail.com', 22, 'Nagpur'),  
-> (5, 'Priya', 'priya@gmail.com', 20, 'Delhi');
```

```
Query OK, 5 rows affected (0.00 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

Query 3.2: Insert Records into Instructor Table

```
mysql> INSERT INTO Instructor VALUES  
-> (101, 'Dr. Patil', 'patil@gmail.com', 'IT'),  
-> (102, 'Dr. Sharma', 'sharma@gmail.com', 'CS');
```

```
Query OK, 2 rows affected (0.01 sec)  
Records: 2 Duplicates: 0 Warnings: 0
```

Query 3.3: Insert Records into Course Table

```
mysql> INSERT INTO Course VALUES  
-> (201, 'DBMS', 4, 101),  
-> (202, 'Data Science', 3, 102);
```

```
Query OK, 2 rows affected (0.00 sec)  
Records: 2 Duplicates: 0 Warnings: 0
```

Query 3.4: Insert Records into Enrollment Table

```
mysql> INSERT INTO Enrollment VALUES
-> (1, 1, 201, '2024-01-10'),
-> (2, 2, 202, '2024-01-11'),
-> (3, 3, 201, '2024-01-12');
```

```
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

4. Display Table Data

Query 4.1: View Student Table

```
mysql> SELECT * FROM Student;
```

```
+-----+-----+-----+-----+-----+
| StudentID | Name   | Email                | Age | Address |
+-----+-----+-----+-----+-----+
|          1 | Ramesh | ramesh@gmail.com    | 20  | Pune    |
|          2 | Amit   | amit@gmail.com       | 21  | Mumbai  |
|          3 | Sneha  | sneha@gmail.com      | 19  | Nashik  |
|          4 | Rahul  | rahul@gmail.com      | 22  | Nagpur  |
|          5 | Priya  | priya@gmail.com      | 20  | Delhi   |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Query 4.2: View Instructor Table

```
mysql> SELECT * FROM Instructor;
```

```
+-----+-----+-----+-----+
| InstructorID | Name      | Email                | Department |
+-----+-----+-----+-----+
|          101 | Dr. Patil | patil@gmail.com      | IT         |
|          102 | Dr. Sharma | sharma@gmail.com     | CS         |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Query 4.3: View Course Table

```
mysql> SELECT * FROM Course;
```

```
+-----+-----+-----+-----+
| CourseID | CourseName | Credits | InstructorID |
+-----+-----+-----+-----+
|        201 | DBMS       |      4 |          101 |
|        202 | Data Science |      3 |          102 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Query 4.4: View Enrollment Table

```
mysql> SELECT * FROM Enrollment;
```

EnrollmentID	StudentID	CourseID	EnrollmentDate
1	1	201	2024-01-10
2	2	202	2024-01-11
3	3	201	2024-01-12

3 rows in set (0.00 sec)